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Cambridge Assessment International Education

Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

8986046942

MATHEMATICS **9709/11**

Paper 1 Pure Mathematics 1 **October/November 2024**

1 hour 50 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

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1 In the expansion of $\left(kx+\frac{2}{x}\right)^4$, where k is a positive constant, the term independent of x is equal to 150.

Find the value of k and hence determine the coefficient of x^2 in the expansion. [4]

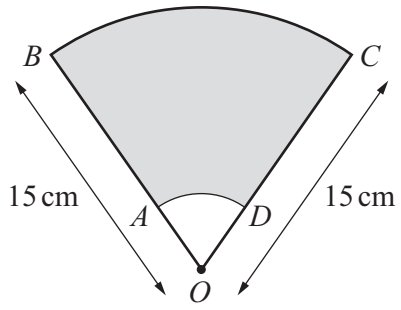
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2 The curve $y = x^2 - \frac{a}{x}$ has a stationary point at $(-3, b)$.

Find the values of the constants a and b. [4]

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The diagram shows a sector of a circle, centre O, where $OB = OC = 15\,\text{cm}$. The size of angle BOC is $\frac{2}{5}\pi$ radians. Points A and D on the lines OB and OC respectively are joined by an arc AD of a circle with centre O. The shaded region is bounded by the arcs AD and BC and by the straight lines AB and DC. It is given that the area of the shaded region is $\frac{209}{5}\pi\,\text{cm}^2$.

Find the perimeter of the shaded region. Give your answer in terms of π. [5]

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4 Show that the curve with equation $x^2 - 3xy - 40 = 0$ and the line with equation $3x + y + k = 0$ meet for all values of the constant k. [5]

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5 The equation of a curve is such that $\frac{dy}{dx} = 4x - 3\sqrt{x} + 1$.

(a) Find the x-coordinate of the point on the curve at which the gradient is $\frac{11}{2}$. [3]

(b) Given that the curve passes through the point (4, 11), find the equation of the curve. [4]

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6 Circles C_1 and C_2 have equations

$$x^2 + y^2 + 6x - 10y + 18 = 0 \text{ and } (x-9)^2 + (y+4)^2 - 64 = 0$$

respectively.

(a) Find the distance between the centres of the circles. [4]

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P and Q are points on C_1 and C_2 respectively. The distance between P and Q is denoted by d.

(b) Find the greatest and least possible values of d. [3]

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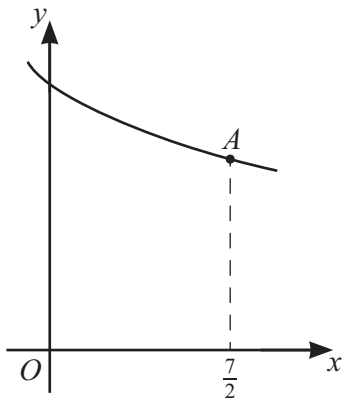
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The diagram shows part of the curve with equation $y = \frac{12}{\sqrt[3]{2x+1}}$. The point A on the curve has coordinates $\left(\frac{7}{2}, 6\right)$.

(a) Find the equation of the tangent to the curve at A. Give your answer in the form $y = mx + c$. [4]

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(b) Find the area of the region bounded by the curve and the lines $x = 0$, $x = \frac{7}{2}$ and $y = 0$. [4]

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8 **(a)** It is given that β is an angle between 90° and 180° such that $\sin\beta = a$.

Express $\tan^2\beta - 3\sin\beta\cos\beta$ in terms of a. [3]

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(b) Solve the equation $\sin^2\theta + 2\cos^2\theta = 4\sin\theta + 3$ for $0° < \theta < 360°$. [5]

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9 The equation of a curve is $y = 4 + 5x + 6x^2 - 3x^3$.

(a) Find the set of values of x for which y decreases as x increases. [4]

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(b) It is given that $y = 9x + k$ is a tangent to the curve.

Find the value of the constant k. [4]

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10 An arithmetic progression has first term 5 and common difference d, where $d > 0$. The second, fifth and eleventh terms of the arithmetic progression, in that order, are the first three terms of a geometric progression.

(a) Find the value of d. [3]

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(b) The sum of the first 77 terms of the arithmetic progression is denoted by S_{77}. The sum of the first 10 terms of the geometric progression is denoted by G_{10}.

Find the value of $S_{77} - G_{10}$. [5]

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11 The function f is defined by $\mathrm{f}(x) = 3 + 6x - 2x^2$ for $x \in \mathbb{R}$.

(a) Express $\mathrm{f}(x)$ in the form $a - b(x - c)^2$, where a, b and c are constants, and state the range of f. [3]

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(b) The graph of $y = \mathrm{f}(x)$ is transformed to the graph of $y = \mathrm{h}(x)$ by a reflection in one of the axes followed by a translation. It is given that the graph of $y = \mathrm{h}(x)$ has a minimum point at the origin.

Give details of the reflection and translation involved. [2]

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The function g is defined by $g(x) = 3 + 6x - 2x^2$ for $x \leqslant 0$.

(c) Sketch the graph of $y = g(x)$ and explain why g is a one-one function. You are **not** required to find the coordinates of any intersections with the axes. [2]

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(d) Sketch the graph of $y = g^{-1}(x)$ on your diagram in **(c)**, and find an expression for $g^{-1}(x)$. You should label the two graphs in your diagram appropriately and show any relevant mirror line. [4]

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Valia

DUBAI CREEK HARBOUR

EMAAR



The Future of *Waterfront Living*

Dubai Creek Harbour is a testament to the future of urban living, where the historic charm of the Creek meets the bold ambition of a modern skyline. It features iconic amenities like Creek Marina, a haven for yachting enthusiasts, and close proximity to the Ras Al Khor Wildlife Sanctuary, renowned for its rich biodiversity, including flamingos.

Iconic Views.
Incredible Access

Positioned at a strategic crossroads, the area offers unparalleled access to the city's most iconic destinations. You are perfectly placed across from a world-class retail and entertainment hub, with the tranquil canal and efficient metro links waiting just on the other side to whisk you across the emirate.



A Life *Well-Connected*

10 Mins to Dubai Intl. Airport

15 Mins to Burj Khalifa / Downtown Dubai

25 Mins to Dubai Marina

40 Mins to Al Maktoum Intl. Airport



Welcome to

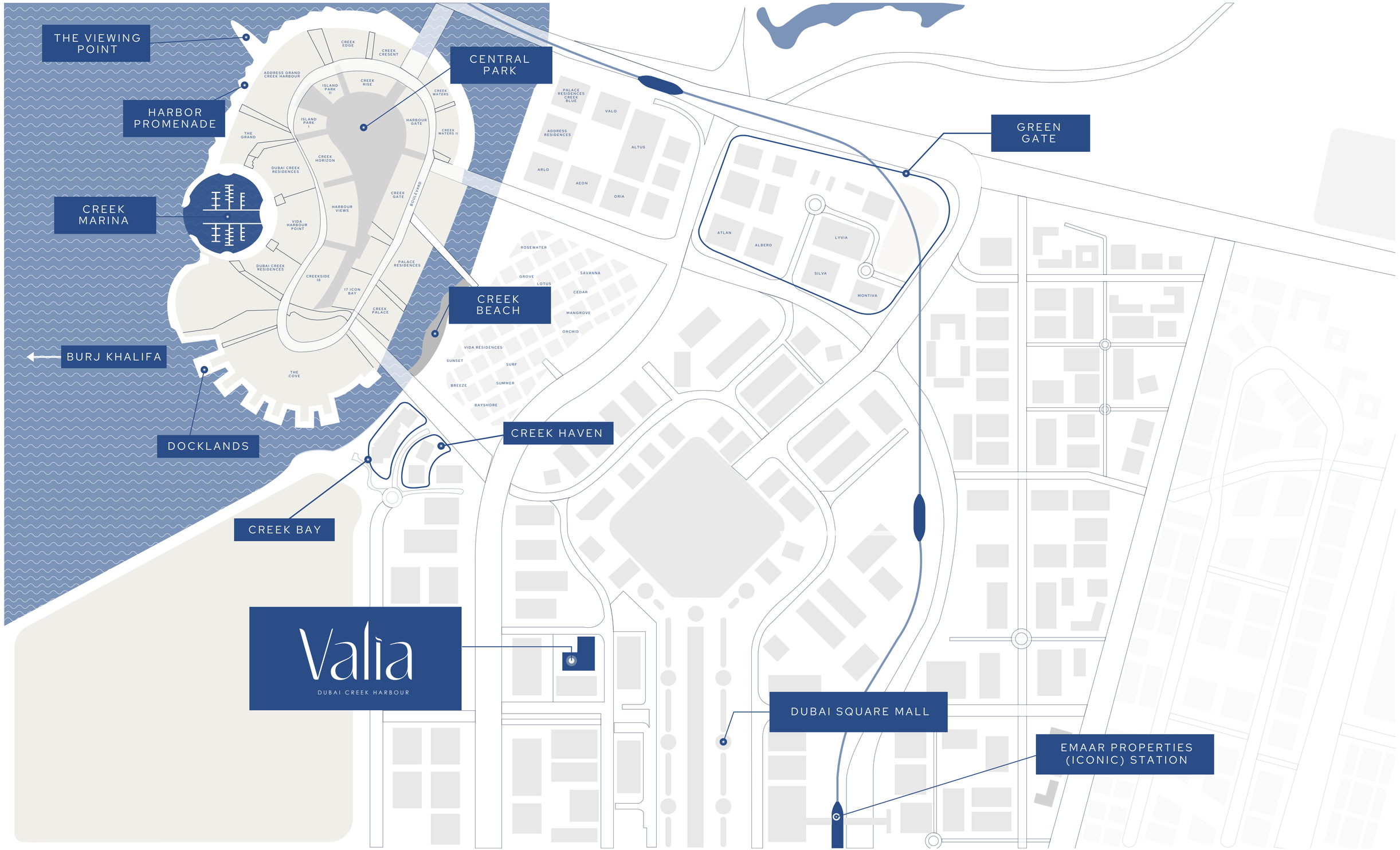
Valia



City Vibrancy, *Naturally Refined*

Valia stands as a sophisticated landmark of contemporary living, a high-rise retreat rising elegantly from a vibrant mixed-use podium. Architecturally sensitive to its surroundings, the project is a sanctuary designed with a profound focus on the well-being of its residents, ensuring every space is an invitation to breathe and belong. Valia creates a residential experience that is as grounded as it is elevated.




THE VIEWING POINT
HARBOR PROMENADE
CREEK MARINA
BURJ KHALIFA
DOCKLANDS
CENTRAL PARK
CREEK BEACH
CREEK HAVEN
CREEK BAY
GREEN GATE
DUBAI SQUARE MALL
EMAAR PROPERTIES (ICONIC) STATION
Valia
DUBAI CREEK HARBOUR
CREEK EDGE
CREEK CRESENT
ADDRESS GRAND CREEK HARBOUR
ISLAND PARK II
CREEK RISE
CREEK WATERS
ISLAND PARK I
HARBOUR GATE
CREEK WATERS II
THE GRAND
CREEK HORIZON
DUBAI CREEK RESIDENCES
CREEK GATE
BOULEVARD
HARBOUR VIEWS
VIDA HARBOUR POINT
PALACE RESIDENCES
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CREEKSIDE 18
17 ICON BAY
CREEK PALACE
THE COVE
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VALO
ADDRESS RESIDENCES
ALTUS
ARLO
AEON
ORIA
ROSEWATER
GROVE
SAVANNA
LOTUS
CEDAR
MANGROVE
ORCHID
VIDA RESIDENCES
SUNSET
SURF
BREEZE
SUMMER
BAYSHORE
ATLAN
ALBERO
LYVIA
SILVA
MONTIVA


LUNE

Your Urban *Pivot*

Nestled within a lively mixed-use neighborhood, this district serves as the ultimate pivot between high-energy urbanism and serene residential life. Here, the pulse of the boulevard meets the quiet rustle of a neighborhood pocket park, creating a community edge where every convenience is just a short stroll away.





A Rhythm of *Well-Being*

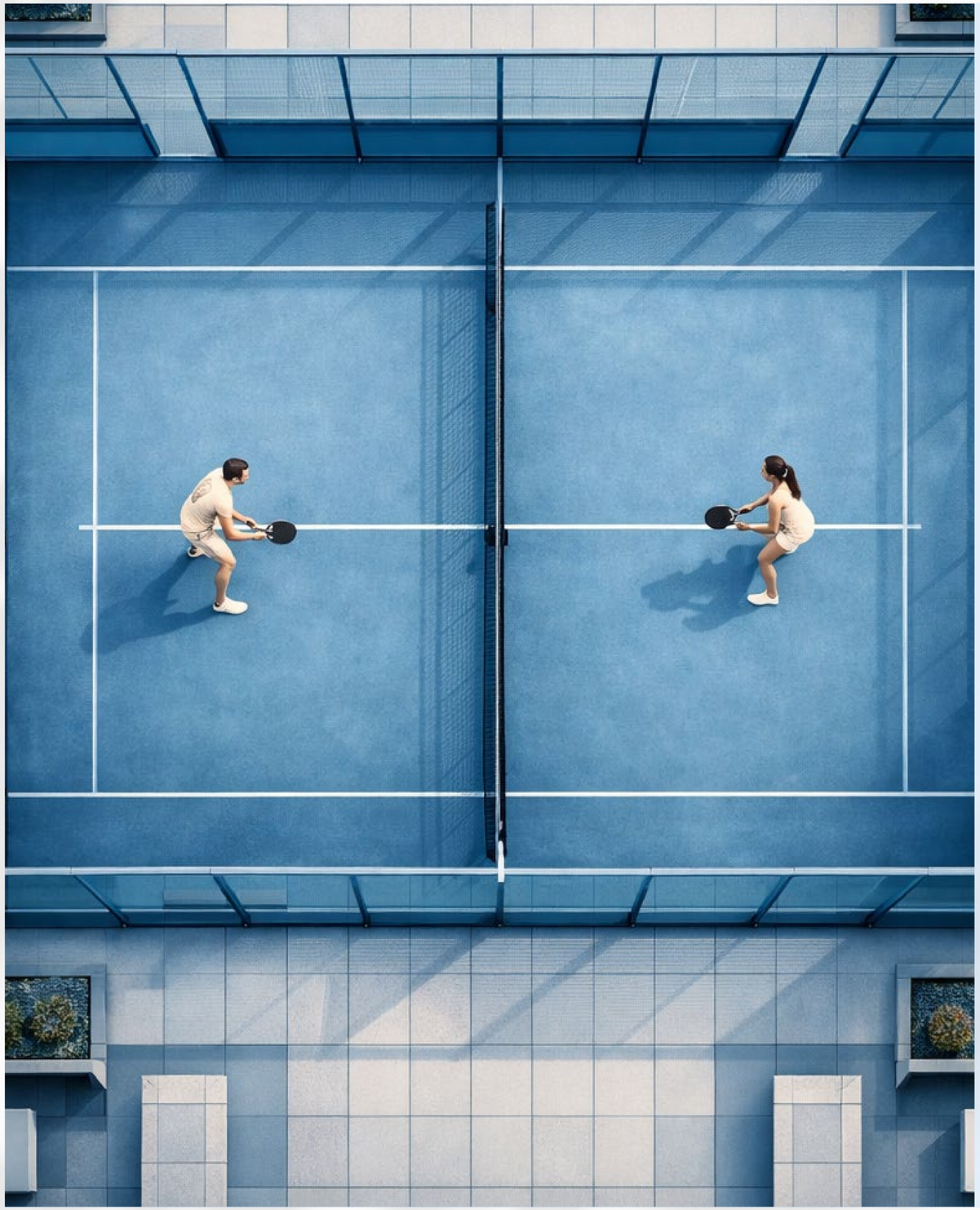
Life at Valia is about finding your own pace in a space designed for connection. Whether you are drawing inspiration from the expansive arrival experience—complete with soothing water features—or enjoying a quiet sunset over the linear park, the lifestyle here is naturally refined and deeply personal.





Thoughtful *Sophistication*

Step into a world of understated luxury, where height and light take center stage. From the grand, high-ceiling lobby that welcomes you home to the meticulously planned layouts, every interior detail is crafted to maximize your sense of space and serenity.



Space to
Breathe & Belong



Relax & Gather

Unwind on the yoga deck, host a sunset feast at the BBQ area, or find shade in a private cabana.



Move & Play

Stay active at the outdoor fitness area and Padel court, while children explore the splash pad and outdoor play zones.



Refresh

Dive into the temperature-controlled swimming pool or watch the little ones in the dedicated kids' pool.

Contemporary *Sensitivity*

The exterior of Valia demonstrates a profound sensitivity to its environment, utilizing authentic, locally sourced materials and delicately proportioned massing. High window glazing and sleek aluminum cladding create a façade that plays with the light, while every unit enjoys its own dedicated balcony to invite the outside in.



Space for *Every Story*

Valia offers a diverse collection of residences tailored to the modern lifestyle, ranging from intimate urban retreats to expansive family sanctuaries. Each home is an exercise in spatial efficiency, designed to cater to the unique needs of its inhabitants while providing a constant connection to the vibrant masterplan outside. Whether you are seeking the simplicity of a one-bedroom apartment or the grandeur of a four-bedroom apartment. Valia provides the perfect canvas for your life.









Crafted for *Living*

A home at Valia is a masterclass in functional elegance, where everyday rituals are elevated by premium finishes and intelligent design. These residences are meticulously prepared for immediate comfort, blending high-end culinary spaces with restful, private quarters that serve as a quiet escape from the city's pulse. Beyond the aesthetic, each home is a testament to sustainable living, integrated with advanced systems that monitor and manage energy with effortless precision.



Bringing City Life
Closer to Home

The upcoming new Blue Line Metro will place Dubai Creek Harbour at the nexus of culture, leisure, and city life. From spontaneous Downtown dinners to breezy beach days, it sets out to bring the best of Dubai within easy reach. This is more than a home; it's your front-row seat to the future of connected urban living, right here in the heart of it all.



Global Lifestyle *Developer*

A leader in the global real estate market, EMAAR has grown over the years to become the largest real estate developer outside of China. You will be investing with one of the best-regarded brands in the world*.

AED 124.2 Bn
Market Cap - FY 2025

AED 25.6 Bn
Net Profit - FY 2025

AED 49.6 Bn*
Revenue

10+
Global Market Presence

200,000+
Residential Units Delivered Globally

53,000+
Residential Units Currently Under Development

*As of 31 st December 2025

Valia

DUBAI CREEK HARBOUR

For more information please call 800 36227 (UAE) / +971 4 366 1688 (International)
or talk to our Property Advisor directly at +971 4888 8844 from 9 AM to 7 PM (GST), Monday to Friday.
Visit our online sales centre at emaar.com or any of our Sales Centres across the UAE.

DUBAI
Emaar Sales Centre | Next to Souq Al Bahar, Downtown Dubai
9:30 AM to 10 PM, Sunday - Friday 9:30 AM to 7 PM, Saturday

Dubai Hills Estate Sales Pavilion | Umm Suqeim Road
Dubai Creek Harbour Sales Pavilion | Ras Al Khor
From 9:30 AM to 7 PM, Sunday - Saturday

EMAAR

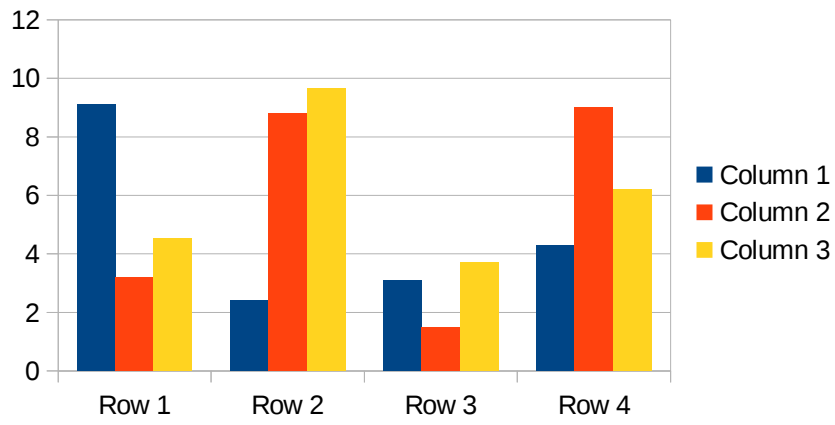
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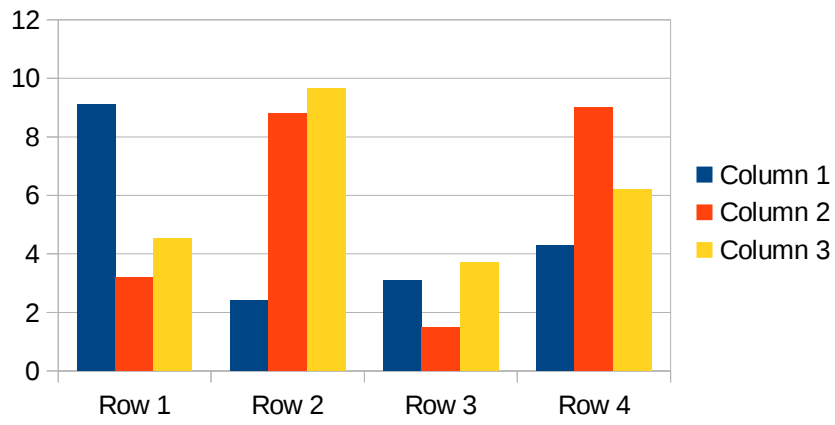
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